

TECHNOLOGY CONSULTING

IN THE GLOBAL COMMUNITY

Final Consulting Report

Palau HCF

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August 2025

Carnegie Mellon University





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I. About the Organizations

A. Overview

The **Republic of Palau Social Security Administration (ROPSSA)** is a self-supporting government corporation headquartered in Koror, Palau. Created by the Social Security Act of 1947 (Title 41, Palau National Code), ROPSSA safeguards the financial well-being of Palauan workers and their families by administering retirement, disability, and survivor benefits. Operations are funded primarily through mandatory payroll contributions, supplemented by investment returns on the Social Security Trust Fund and periodic national-government subsidies. A five-member Board of Trustees provides strategic oversight, while Administrator Clinton O. Ngemaes directs day-to-day activity.

Under the same statutory framework, ROPSSA also administers the **Republic of Palau Healthcare Fund (HCF)**—the nation’s first universal healthcare-financing system—established by the National Healthcare Financing Act of 2010 (RPPL 8-14). HCF pursues universal health-coverage goals through two integrated programs: (1) **Medical Savings Accounts (MSA)**, which allow individuals to accumulate pre-tax savings for routine care, and (2) a pooled **National Health Insurance (NHI)** fund that protects households against high-cost or off-island treatment. Like Social Security, HCF is financed by compulsory employer-and-employee contributions collected alongside payroll taxes. Currently, ROPSSA and HCF share a unified organizational structure, with the HCF operating as a subsidiary under ROPSSA. This arrangement is expected to change in the near future with HCF splitting off as a more independent entity.

Together the organizations employ **approximately 35 staff** and serve a constituency of about **2,000 employers, 12,000 active contributors, and 4,000 benefit recipients**.



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B. Mission Statements

(a) The Mission of ROPSSA:

“To ensure eligible persons receive security during old age, disability, or mortality, providing protection for survivors and enhancing community well-being.”

(b) The Mission of HCF:

“To promote the health and social welfare of the citizens of the Republic of Palau through the establishment of a healthcare financing system that provides free or subsidized healthcare for its citizens”

C. Facilities

(a) ROPSSA Temporary Location:

The ROPSSA facility is a temporary, prefabricated building centrally located in Koror. ROPSSA plans to relocate to a new, permanent location in Koror, specifically to the One-Stop-Shop facility (detailed below). Currently, the ROPSSA has cubicle workstations and a sufficient number of computers for its employees. The information technology (IT) team has an office in a separate room with a single server stack. The ROPSSA employees use desktop computers.

(b) Healthcare Fund Building:

HCF is housed separately from the ROPSSA in a building approximately 200 meters away. The HCF has cubicle workstations and computers for its employees, with a well furnished and equipped conference room.

(c) One-Stop-Shop:

The ROPSSA plans to move later this summer into the Palau One-Stop-Shop (OSS), a newly constructed federal government office building that houses the executive office of



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the President along with several ministries including the Ministry of Human Resources, Culture, Tourism & Development and Foreign Investment Board. OSS offers the same IT equipment, team, and LAN server as the temporary ROPSSA office.

D. Programs

ROPSSA administers two key programs fundamental to its mission of safeguarding the financial and social well-being of Palauan workers and their families: the **Social Security Program** and the **Healthcare Fund (HCF)**. These programs directly advance ROPSSA's mission to provide financial security during old age, disability, or in the event of mortality (via survivor benefits), and to support health and social welfare through equitable healthcare financing.

All employees working in Palau are required to contribute to these programs. Employees contribute **7%** of their gross income to the **Social Security Fund** and **2.5%** to the **HCF**. Employers match these contributions, bringing the total combined contribution to **14% for Social Security** and **5% for HCF**. These mandatory contributions are collected quarterly and directly fund benefits for retirees, survivors, and individuals facing health challenges.

Beyond these core contributions, ROPSSA also manages **Medical Savings Accounts (MSAs)**. These accounts enable individuals to set aside pre-tax funds for routine care at local hospitals and medical clinics, supporting the community's access to essential health services and aligning with HCF's mission to promote equitable, sustainable healthcare financing.

Prior to our project work, ROPSSA's activities under these programs relied heavily on manual, paper-based processes. Employers and individuals typically submitted forms and payments in person, with staff re-entering this data into proprietary software systems. These processes led to redundancy and a high volume of paper records stored in multiple locations. Staff often faced challenges in keeping track of form updates and ensuring accurate record-keeping across paper and digital systems..



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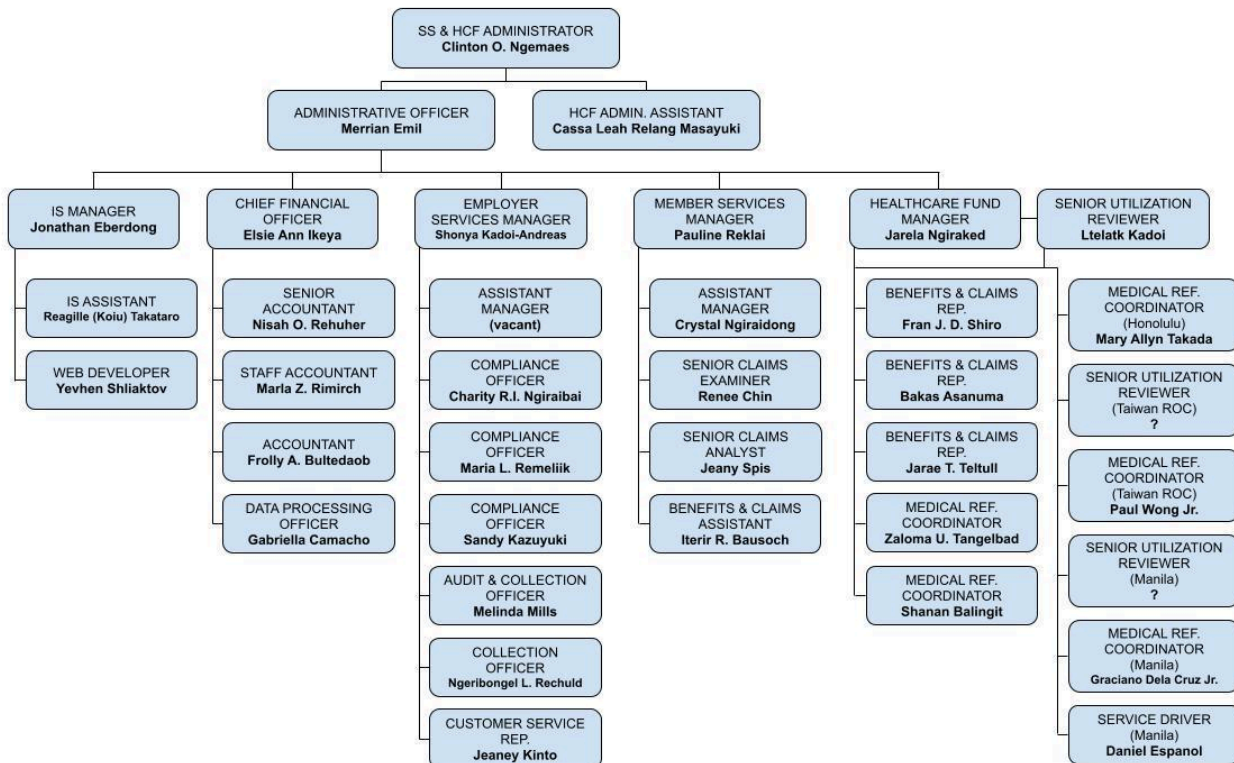
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E. Staff

The organizations employ **approximately 35 staff**—23 assigned to Social Security functions and 12 dedicated to HCF administration. Below is an overview of the structure of employees at ROPSSA and HCF as of May 2025:



The administrative team, our main point of contact for this project, includes the following personnel:

(a) **Merrian Emil, Administrative Officer:**



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Merrian provides comprehensive administrative support to the various managers and staff at ROPSSA. She is heavily involved in overseeing office operations and logistics and managing records.

(b) Pauline Reklai, Members Service Manager:

Pauline oversees all aspects of member services, ensuring efficient service delivery and handling member inquiries and support needs. She works closely with other departments to ensure members receive accurate and timely assistance.

(c) Shonya Kadol-Andreas, Employer Services Manager:

Shonya manages employer-related processes and services, including coordinating employer reporting and contribution compliance. She ensures employers have the information and support needed to meet their obligations under ROPSSA and HCF programs.

Overall, SSA and HCF staff are proficient in using Microsoft Office tools, their Dell workstations, and Windows software to complete their daily operations. Assistance and training are provided by the IS team. This technology consulting project, initiated by Administrative Officer Merrian Emil and IS Manager Jonathan Eberdong, primarily impacts the Employee Services Section (ESS), Member Services Section (MSS), and HCF management.

The information systems (IS) team includes the following personnel:

(a) Jonathan Eberdong, IS Manager:

Jonathan is a Computer Hardware Engineer. He manages all of the organization's technology management and planning needs.

(b) Reagille (Koiu) Takataro, IS Assistant:

Reagille supports the IS team in responding to requests and updates.

(c) Yevhen (Genya) Shliaktov, Web Developer:

Yevhen manages the ROPSSA website and assists in responding to requests and updates.



F. Technology Infrastructure

As of the project's inception, ROPSSA's technology infrastructure comprised a **hybrid digital-paper environment**, characterized by older yet generally functional hardware coexisting with a heavy reliance on paper-based forms and redundant manual processes.

Most forms—including contribution paperwork, benefit applications, and internal forms—were submitted on paper, or paper-based copies were maintained even when digital versions existed. This redundancy contributed to confusion, multiple data transcriptions, and inconsistent updates across systems.

Overall, while computers and digital tools like Microsoft Office are central to daily administrative tasks, staff still relied heavily on manual processes to ensure compliance and recordkeeping.

Technology Infrastructure	
Hardware (temporary office)	Dell Workstations: Dell OptiPlex SFF 5050 Intel Core i5 Servers: Dell PowerEdge R720 Server
Hardware (One-Stop-Shop)	Dell Workstations: Dell OptiPlex SFF Plus 7020 CPU: Intel Core i7, 2.10 GHz GPU: Intel UHD Graphics 32 GB Ram Servers: Dell PowerEdge R720 Server
Network	TCP/IP LAN Network
Internet Connection (temporary office and One-Stop-Shop)	32 MBPS Shared Connection
Threat Management	Untangle firewall and web filters



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OS Server	Windows 2022 SQL 2022
OS WorkStation (temporary office)	Windows 10 Pro
OS WorkStation (One-Stop-Shop)	Windows 11 Pro 64-bit
Software Programs	Microsoft Office Package 2016 (temporary office) Microsoft Office 2022 (OSS) ROPSSA/HCF OSS system

While slightly outdated, the current operating systems and workstations remain functional and practical for employees, with only occasional issues arising at the individual workstation level. ROPSSA is also expected to have a small variety of hardware upgrades at their new location, the One-Stop-Shop. This provides a unique opportunity to transition to more modern software alternatives that may support more robust IT solutions and make services more efficient and organized.

Internet is provided by two main providers, the **Palau National Communications Corporation (PNCC)** and **Palau Telecommunications**. It is often unreliable and slow; however, this is a pervasive issue across all internet connections on the island, thus lacking a readily available solution. This limits ROPSSA/HCF to the use of local networks, as over-reliance on the cloud could lead to difficulties accessing crucial data. Members of the IT team have also expressed concern over security issues related to switching to a web-based server and are understandably reluctant to pay for an annual subscription for a cloud-based service, citing their current reliance on and satisfaction with the existing system.

G. Technology Management

As noted in the staff section above, technology at ROPSSA is managed internally by the IS team. The IS team is responsible for maintaining **all hardware, software, networking, and application components**. They address problems as they arise, typically reported by staff



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through email, SMS, or in-person requests. A basic Excel-based log exists for tracking issues, but its infrequent updates and informal usage limit its effectiveness.

Day-to-day tasks include:

- (a) Resolving device-level or network problems internally.
- (b) Managing the ROPSSA website (daily updates handled by the Web Developer).
- (c) Managing daily automated software backups and updates across workstations and servers.

For more advanced or large-scale infrastructure projects, ROPSSA has historically contracted with off-island vendors, relying on external expertise for major improvements. This reliance is partially mitigated by maintaining multiple redundant servers and hardware to ensure operations continue even when external help is delayed.

H. Technology Planning

The IS team is also responsible for identifying and proposing technology infrastructure projects. However, any project requires the Administrator's authorization before proceeding, and significant budget requests are also reviewed with the Board of Trustees. The budget itself is managed by the Chief Financial Officer (CFO), Elsie Ann Ikeya.

Technology planning has traditionally been ad hoc, with no formal long-term technology strategy in place. Most infrastructure upgrades have been reactive, focused on maintaining continuity of operations (e.g., ensuring backups are reliable, servers are functional).

One past IS project involved implementing a secure process for submitting forms and payments online. This initiative leveraged the expertise of a student consultant from **Carnegie Mellon University's Tech Consulting in the Global Community (TCinGC) program**, the same program supporting the current project. More information about that project can be found [here](#). Day-to-day hardware and software maintenance is typically managed internally by the IS team.



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At the start of our project, there was no comprehensive technology plan guiding broader modernization or digital transformation efforts. This gap provided a key opportunity to propose a more coordinated, future-proof technology plan as part of our project deliverables.

I. Information Management

As previously noted, information flow among staff typically occurs via paper, with pertinent data (such as employee forms) requiring duplicate entry into the software. They have not attempted to automate this in the past. Shortcomings of the current system include the following:

- (a) Multiple data transcriptions into software (duplication of work)
- (b) Little to no automation (operations by paper)
- (c) Redundant information (see below)

Manuals, forms, and other paperwork are currently stored in three different locations:

- (a) Paper copies stored in binders in the administrative section of the temporary office
- (b) On a shared local network drive
- (c) Backed up on hard drives of respective managers and administrators

With the passing of amendments, administrative changes, and evolving laws, numerous updates are required for forms and manuals, necessitating their modification across all three storage locations. The lack of a unified information repository leads to avoidable confusion and inconsistencies across the organization. Employee manuals, operations manuals, and forms are critical to ensure fair, ethical, and compliant work processes. Therefore, it is essential that these manuals be accurate, up-to-date, and easily accessible to managers and their support staff.



II. Modernization and Consolidation of ROPSSA and HCF Employee Manual

A. Opportunity

Prior to our work, ROPSSA and HCF relied on **three distinct employee manuals** from **2013**, **2018**, and **2023**. Each version contained outdated, contradictory, or incomplete policies and language that did not reflect the current operational practices of the two organizations. This created confusion among staff, inconsistencies in supervisory decisions, and inefficiencies in onboarding and day-to-day work. These inconsistencies undermined the organizations' ability to uphold fair and consistent internal practices and to present a unified set of employee guidelines.

In the absence of a consolidated and regularly updated manual, managerial personnel relied heavily on personal recollection to communicate the prevailing employee policies. This reliance on memory, paired with inconsistent documentation, introduced significant risks of policy misinterpretation and operational inconsistencies.

As HCF is expected to soon become more independent from ROPSSA, an additional opportunity arose to ensure the shared employee manual's language explicitly acknowledged and supported both entities while maintaining clarity and relevance for their evolving structures. Additionally, while the initial objective was to consolidate and update existing policies, it became clear during the review process that a more comprehensive rewrite was needed to address not only policy inconsistencies but also pervasive grammatical errors, outdated language, and structural disorganization that hampered usability and staff understanding.

B. Why This Matters

Updating and consolidating the shared employee manual is critical for several reasons:



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(a) Clarity and Consistency:

Without a single, accurate, and up-to-date manual, staff rely heavily on personal knowledge or informal interpretations, which risks inconsistent application of policies and confusion among employees and managers.

(b) Onboarding and Staff Support:

A modern, well-organized employee manual streamlines onboarding and daily work, reducing the burden on supervisors to clarify basic procedures and improving staff confidence in following procedures.

(c) Efficiency and Professionalism:

The outdated and contradictory manuals slow down internal processes and create avoidable redundancies. A rewritten manual, accessible digitally and designed for ease of navigation, empowers staff to quickly find accurate information, supporting smoother operations across departments.

(d) Future-Proofing:

As HCF becomes a more independent entity, a unified and clear employee manual that appropriately reflects the roles and policies applicable to both ROPSSA and HCF ensures a smooth transition and maintains operational coherence in the future.

C. Alternative Solutions Considered

Option	Evaluation	Drawbacks
Maintain the Status Quo	Retaining the most recent manual and making no changes.	This would have left staff with a document full of errors, outdated content, and contradictory information,



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		perpetuating confusion and inefficiencies.
Limited Edits	Making minor updates to the 2023 manual, accounting for broad policy changes, retaining the organizational structure and language whenever possible.	This would not have addressed the foundational issues in structure, organization, and outdated content that were uncovered during our preliminary review of the manuals.
Proposed Solution: Comprehensive Rewrite	Fully rewriting the manual, consolidating accurate policies from all three versions, eliminating contradictions, modernizing the language, and reorganizing it for clarity and usability.	This approach is more time-intensive but ensures the final product is tailored to the real working practices of ROPSSA and HCF and meets the organizations' current and future needs.

D. Activities

- (a) **Conducted a comprehensive review** of the 2013, 2018, and 2023 employee manuals to identify contradictions, outdated policies, and grammar issues.
- (b) **Met with ROPSSA and HCF leadership** (including Merrian Emil, Shonya Kadol-Andreas, and Pauline Reklai) as needed to ensure policies are up-to-date and accurately reflect current operations.
- (c) **Rewrote the manual** in plain, professional language, reorganizing it to improve structure and usability.



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- (d) **Developed a digital version of the manual** online using Google Sites, with policies compartmentalized into pages for easy navigation and searchability.
- (e) **Created digital versions of paper-based forms** using Google Forms and embed them within the Google Sites platform.

E. Expected Outcomes

Outcome Area	Baseline	Target	Measurement Approach
Policy clarity and consistency	Frequent confusion and reliance on memory	Unified, clear policies applicable to both entities	Post-launch staff survey and supervisor feedback
New-hire onboarding effectiveness	Onboarding process inconsistent and manual-heavy	Clear, consistent onboarding supported by digital access	Qualitative interviews with staff post-launch

F. Outputs

Deliverable	Description
Fully updated employee manual	Comprehensive rewrite of the employee manual consolidating accurate policies, using professional, modern language, and ensuring clarity throughout.
Digital, internet-based manual platform	Custom Google Sites website with the manual organized into searchable, clearly categorized sections.



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Updated Google Forms for staff use	Transition of previously paper-based forms (leave requests, travel requests, etc.) into easy-to-use, secure digital forms.
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G. Feasibility

The project was feasible within the given timeframe and resources. The ROPSSA and HCF leadership team strongly supported the work and provided critical feedback throughout. The IS team has the technical capability to support the Google Sites site and update the digital forms as needed.

H. Sustainability

To ensure long-term sustainability, the project includes:

(a) IS Team Ownership:

Responsibility for future updates is assigned to the Employee Services Manager and the IS team.

(b) Comprehensive Training:

We provided thorough training for staff on Google Sites usage and on accessing and updating the manual.

(c) Enhanced Ability and Control:

Staff have the option to use a third-party site traverser to host the site locally through IIS, providing an additional layer of control and accessibility if desired.



III. Development of Distinct Operations Manuals for ROPSSA and HCF

A. Opportunity

ROPSSA and HCF currently share a unified organizational structure and historically relied on a single operations manual. This manual had become outdated and did not reflect the actual day-to-day practices or unique operational realities of the two entities. The manual contained redundant information, processes no longer in use, and lacked clarity around how operations differ between the two organizations. Moreover, as HCF plans to become a more independent entity from ROPSSA in the near future, there was an urgent need for each organization to have its own tailored operations manual that accurately reflects current processes, future governance models, and the roles of staff and leadership.

Unlike the employee manual, the operations manuals held direct and severe legal implications because they were required to be consistent with the statutory obligations of ROPSSA under Title 41 of the Palau National Code, which sets out the duties, governance, and operating parameters of the Social Security Administration and related entities. Operating without clear, accurate, and legally aligned manuals increased the risk of operational inefficiencies, confusion about organizational mandates, and potential non-compliance with Palau's statutory framework.

B. Why This Matters

(a) Organizational Readiness for HCF's Independence:

With HCF moving toward independence, having distinct and accurate operations manuals is essential to clarify roles, processes, and responsibilities that will ensure a smooth transition and continued operational excellence.



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(b) Compliance and Legal Risk Mitigation:

Title 41 of the PNC directly governs the work of ROPSSA and its relationship to national healthcare financing efforts through HCF. Manuals that are inaccurate or fail to reflect the requirements of the PNC expose the organizations to legal risk, financial penalties, and reputational damage.

(c) Operational Efficiency and Clarity:

Clear, distinct manuals for each organization ensure that staff can confidently execute their roles, reducing the need for ad hoc decision-making and reliance on institutional memory, which can lead to errors and inconsistencies.

(d) Accountability and Transparency:

Updated manuals enhance transparency in decision-making processes and support the broader mission of ROPSSA and HCF to serve Palauan citizens equitably and efficiently.

C. Alternative Solutions Considered

Option	Evaluation	Drawbacks
Maintain a Shared Manual	Would maintain some continuity across entities, but fails to reflect operational divergence.	Would not address the unique processes of each organization, creating future inefficiencies.
Partial Edits to the Current Manual	Faster and less labor-intensive, but does not address the full scope of outdated and duplicate content.	Leaves persistent ambiguities and risks of legal non-compliance with Title 41.



Proposed Solution: Full Rewrite of Separate Manuals	Comprehensive approach that ensures each manual aligns with the relevant legal, operational, and organizational realities.	More time-intensive but necessary to ensure future-proof clarity and compliance.
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Given the imminent structural changes and the complexities of Title 41 compliance, the full rewrite of distinct manuals for each entity was deemed the only viable path to ensure accuracy, legality, and operational effectiveness.

D. Activities

- (a) Conducted a **gap analysis** of the existing shared manual, identifying outdated, conflicting, or duplicative sections that do not reflect Title 41 or current organizational practices.
- (b) **Collaborated with managers** across departments—including Administrative, Employee Services, and Member Services—to capture current practices and operational pain points.
- (c) **Drafted separate, plain-language manuals** for ROPSSA and HCF, ensuring that each reflects its specific governance structure, operational processes, and future independence.
- (d) Uploaded the new manuals to the **Google Sites website** for easy access, security, and navigational structures to support staff adoption.
- (e) Provided **training and handover sessions** for staff and managers to ensure they can locate, understand, and update the manuals in the future.

E. Outcomes



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Outcome Area	Baseline Situation	Target Outcome	Measurement Approach
Legal alignment of manuals	Outdated content; partial alignment with Title 41	Full compliance with Title 41, clarity in roles and processes	Sign-off by the ROPSSA Administrator
Operational clarity	Staff confusion on internal processes	Clear, step-by-step procedures for both entities	Supervisor and staff feedback
Preparation for HCF split	Shared manual structure, no clear delineation	Two distinct manuals reflecting separate governance and functions	Supervisor and staff feedback

F. Outputs

Deliverable	Description
Two distinct operations manuals (ROPSSA & HCF)	Comprehensive, plain-language documents outlining specific roles, responsibilities, and processes unique to each organization.
Digital versions accessible on the internet	Hosted on the same Google Sites as the employee manual for easy, secure access.

G. Feasibility



This work was feasible within the project timeline and received strong support from both ROPSSA and HCF leadership, who were committed to ensuring that their staff have accurate, up-to-date operational guidance. The IS team has the technical ability to support a live website. While rewriting manuals of this scope required close coordination with managers and legal counsel, the foundational work done during the employee manual overhaul positioned us to tackle this next phase.

H. Sustainability

(a) Governance for Updates:

Each operations manual has a designated policy owner (Administrative Officer for ROPSSA, HCF Manager for HCF) responsible for annual reviews and updates.

(b) Flexibility:

Manuals are stored and maintained on a Google Sites website, with the option to host the site locally through IIS.

(c) Skill Transfer:

Managers received training on how to edit and maintain the manuals, ensuring long-term continuity and accuracy.

IV. AI-Powered Legal Compliance and Text Synthesis Tool

A. Opportunity



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ROPSSA and HCF maintained a wide array of internal documents—such as the operations manuals, employee policies, and customer-facing forms—that needed to be accurate, up-to-date, and legally aligned. Historically, updates to these materials were performed manually, requiring significant time from staff and leading to inconsistencies, incomplete updates, and delayed responses to regulatory or operational changes.

Given the complexity of compliance (particularly for operations manuals under Title 41 of the Palau National Code) and the anticipated future divergence of HCF, there was a clear need for a more streamlined, automated way to:

- (a) Analyze legal compliance across internal documents.
- (b) Suggest language updates or content improvements where needed.
- (c) Support efficient document management in a secure, local environment.

This created an opportunity to leverage modern AI technologies—specifically **retrieval-augmented generation (RAG)** and **large language models (LLMs)**—to automate parts of the compliance-checking and drafting process.

B. Why This Matters

(a) Accuracy and Consistency:

Automating compliance checks and language improvements reduces human error and ensures that documents consistently meet legal and operational requirements.

(b) Time Saving:

Staff currently spend significant time manually cross-checking policies and updating language—an automated tool can dramatically reduce that workload.

(c) Future-Proofing:



As HCF becomes more independent and ROPSSA operations continue to evolve, having a tool that can rapidly synthesize new language and check legal alignment will help these agencies remain nimble and compliant.

(d) Empowering the IS Team:

By integrating this tool into the local environment and training the IS team on its use, the agencies can maintain independence from external consultants or costly SaaS solutions.

C. Alternative Solutions Considered

Option	Evaluation	Drawbacks
Continue Manual Updates	Maintains status quo; no additional costs.	High staff time, ongoing inconsistencies, high risk of missed updates.
Purchase Third-Party SaaS Tool	Could offer built-in compliance checks and drafting features.	Ongoing subscription costs, reliance on external vendors, data security risks, internet reliability concerns.
Proposed Solution: In-House Tool Development	Custom-built tool leveraging existing staff skills and AI technologies to provide automated compliance and drafting support.	One-time development effort; requires internal capacity building for the IS team to maintain.

The decision to develop an in-house tool was driven by a desire to ensure data security, align the tool closely with Palauan law and agency practices, and avoid the limitations of internet-based services.



D. Completed Activities

- (a) Designed a roadmap and the technical architecture for the tool, considering the existing hardware, software environment (Windows servers, Dell workstations), and local network constraints.
- (b) Developed AI-powered compliance tool using:
 - (i) **Large language models (LLMs)** for natural language understanding and drafting support.
 - (ii) **LangChain** for integrating data retrieval and semantic search capabilities.
 - (iii) **RAG pipelines** to connect local data with the AI models and produce context-aware suggestions.
- (c) Tested the tool on sample internal documents (e.g., sections of the operations manual and employee manual) to validate accuracy and relevance of the suggestions.
- (d) Developed a **GUI** and created an **installer package** for ease of use for nontechnical staff
- (e) Created **training materials and interactive workshops** for the IS team, ensuring they are comfortable with tool maintenance, updates, and troubleshooting.
- (f) Deployed the tool to the local server environment to ensure easy access and security.
- (g) Collected feedback from staff (the IS team, AO, MS, and ES) to refine the tool's features and output quality.

E. Outcomes

Outcome Area	Current Situation	Target Outcome	Measurement Approach
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Efficiency of manual updates	Slow, error-prone process relying on human input	Automated initial review and suggestions for updates	Staff surveys post-pilot
IS team readiness to maintain tool	No in-house AI tools or knowledge.	IS team fully trained to manage, update, and adapt tool	Completion of training; IS team feedback
Compliance with evolving legal standards	Reactive updates only when gaps are manually identified	Proactive identification of compliance gaps	Audit of tool's flagged updates vs. manual legal review results

F. Outputs

Deliverable	Description
AI-powered compliance tool	Two locally hosted tools that use RAG and LLM to scan internal documents for legal alignment and propose updates.
AI-powered drafting tool	A locally hosted tool that uses LLM to analyze compliance results and suggest edits for clarity.
GUI for tools	Developed a Graphic User Interface for the AI-powered tools for accessibility to staff with less technical expertise
Training modules for IS team	Step-by-step documentation and interactive training sessions to build in-house capacity for using and maintaining the tool.
Deployment on local network	Integration of the tool into the current LAN environment, ensuring accessibility and security.



Version history and change logs	Built-in tracking of edits and updates for transparency and accountability.
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G. Feasibility

The development of this tool was **ambitious** but **feasible**. The IS team supported the digitization of manuals and forms and also confirmed their ability to maintain such a tool. Based on our trials during development, we became confident that the tool could be used on the office's new Dell Workstations and developed it as a **downloadable script** runnable with the application **LM Studio**.

For successful implementation, we drafted a clear integration strategy for the tool with existing systems and secured **commitment from IS team members** for ongoing use and upkeep of the project.

H. Sustainability

(a) Local Hosting:

The tool is hosted on the organizations' local machines, reducing concerns about data security and ensuring resilience even during internet disruptions.

(b) IS Team Ownership:

The project included comprehensive training and documentation so that the IS team can update, maintain, and expand the tool independently.

(c) Change Logs and Governance:

All updates and changes proposed by the tool are logged and reviewed by managers to ensure oversight and ongoing legal compliance.



(d) Alignment with Continuous Improvement Culture:

The tool's development directly supports ROPSSA's stated goal of "continuous improvement" and better service to Palauan citizens.

V. Recommendations for Internal Workflow Modernization

A. Opportunity

At the start of our project, ROPSSA and HCF managed internal workflows using predominantly paper-based processes, legacy software (Office 2016 and InfoPath), with all operations processed on a local area network (LAN). With planned organizational changes—including the impending independence of HCF and relocation to the new One-Stop-Shop building—there existed a unique opportunity to modernize operations and software, streamline workflows, and significantly enhance productivity, accuracy, and efficiency.

Recognizing these issues, we conducted a comprehensive **process audit** to determine a **technology modernization strategy**. Our aim was to recommend solutions that:

- (a) Improve efficiency and accuracy of data management.
- (b) Enhance staff productivity by reducing manual and duplicative tasks.
- (c) Ensure sustainability within the existing IT environment.
- (d) Support future transitions, such as the independence of HCF and the move to the One-Stop-Shop facility.



B. Why This Matters

(a) Operational Efficiency:

By reducing duplicative data entry and automating processes, staff time can be redirected to mission-critical work serving Palauan citizens.

(b) Security and Compliance:

Moving away from scattered paper-based forms and shared folder silos enhance document security and reduce the risk of sensitive information mishandling.

(c) Staff Satisfaction:

A modern, user-friendly system minimizes frustration and error-prone workarounds, directly contributing to morale and staff retention.

(d) Sustainability and Future Readiness:

With ROPSSA/HCF transitioning to new physical and organizational structures, adopting sustainable digital tools will ensure a seamless future.

C. Completed Activities

During our engagement, we:

- (a) Evaluated existing processes, including the reliance on InfoPath (which was slated for discontinuation), Microsoft Office 2016 (with support ending October 2025), and the aging Dell hardware and LAN infrastructure.



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- (b) Conducted interviews with IS manager Jonathan Eberdong about current software and hardware environment (see Technology Infrastructure). The most critical insights were:
- (i) ROPSSA planned to transition software packages from Microsoft Office 2016 to Office 2022.
 - (ii) They planned to upgrade desktops to Dell OptiPlex SSF Plus 7020 with 32 GB of RAM and operating systems from Windows 10 to Windows 11 Pro 64-bit
 - (iii) The office operated completely via a LAN server out of concerns over security, internet reliability, cost efficiency, and convenience.
- (c) Consulted with administrative leadership (Administrative Officer Merrian Emil, Employee Services Manager Shonya Kadol-Andreas, and Member Services Manager Pauline Reklai) to identify software upgrades and features that can best support their operations. Some of their expressed needs were:
- (i) A streamlined process for submitting forms online to improve operational efficiency and reduce reliance on paper-based forms.
 - (ii) A digitized method of accessing manuals, forms, and other documents. This could be cloud based or optionally integrated into the LAN.
- (d) Identified key pain points, including time-consuming manual workflows, difficulties locating up-to-date documents, and staff concerns about transitioning to cloud-based systems given Palau's bandwidth limitations.
- (e) Compiled a list of potential software solutions to address these issues and aligned them with the agency's operational and technical environment.



D. Decision Matrix: Evaluation of Workflow Modernization Solutions

Criteria	Weight	Rationale
Security	25%	Must protect sensitive HR and operations data; top priority given staff concerns.
Internet Independence	20%	Solutions must work well despite Palau's limited internet connectivity.
Ease of Use	15%	Staff proficiency with Microsoft Office; user-friendly tools essential for adoption.
Longevity/Support	10%	Preference for supported software that will not become obsolete quickly.
Technological Integration	10%	Tech-forward solution for seamless data flow and compatibility with current hardware and software tools.
Cost/Financial Feasibility	20%	Budget-conscious approach needed to avoid large recurring expenses.

Criteria (Weight)	Status Quo (Office 2022 only)	Microsoft 365 (Full Cloud)	Hybrid (LAN Website + Cloud Forms)
Security (25%)	4 (1.0)	4 (1.0)	4 (1.0)



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Internet Independence (20%)	5 (1.0)	2 (0.4)	4 (0.8)
Ease of Use (15%)	4 (0.6)	4 (0.6)	5 (0.75)
Longevity/Support (10%)	4 (0.4)	5 (0.5)	4 (0.4)
Integration (10%)	3 (0.3)	5 (0.5)	4 (0.4)
Cost-effectiveness (20%)	5 (1.0)	1 (0.2)	1 (0.2)
Total Score	4.2	3.2	3.55

*We scored each solution on a 1–5 scale (1 = poor fit, 5 = excellent fit), multiplied by the weight, and summed for an overall weighted score.

E. Recommendation: Optimized LAN-Based Digital Environment

The **Status Quo Option**: The use of a LAN-based system integrating a free, locally-hosted website balanced security, efficiency, internet independence, integration, and long-term scalability. Google sites allows for both a cloud based manual access with the flexibility to be hosted locally, along with cloud-based online form submissions. This strategy offers a clear upgrade path without immediate, full cloud dependency.

Given ROPSSA's recent purchase of Office 2022, it became clear that additionally purchasing an annual Office 365 subscription that offers minimal new features for staff would waste over a thousand dollars per year just to ensure our earlier solutions were compatible via Microsoft-exclusive tools. Consequently, after evaluating all of the service packages, we



recommend ROPSSA to **delay transitioning to cloud-based services** and reevaluate the need for them in 3-5 years.

F. Outcomes

(a) Reduced Manual Handling:

We achieved a significant reduction in manual data handling and redundancy

(b) Enhanced Productivity:

Staff satisfaction and productivity improved thanks to modern interfaces and easy-to-use digital forms.

(c) Tailored Solutions:

Our solutions are tailored to ROPSSA's current and planned software upgrades and hardware environment.

G. Outputs

Deliverable	Description
Website Expansion (Google Sites)	A digital home for employee and operations manuals, complete with categorized tabs and integrated digital forms.
Digital Form Conversion	Convert critical internal forms to Google Forms, significantly reducing manual work and paper waste.
IS Team Training	In-depth sessions on maintaining the digital environment, editing website, and supporting staff use.



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Sustainability Plan for Future Tech Refreshes	Guidelines for maintaining these systems and evaluating future upgrades such as transitioning to Microsoft 365
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H. Feasibility:

(a) Budget:

The upfront licensing for Microsoft Office 2022 is already purchased for 29 computers, which makes the other two options financially inefficient and redundant.

(b) Internet Resilience:

While Google Site is an online platform, its on-premise deployment option for ROPSSA on the LAN ensures that workflows continue to function seamlessly, even during internet disruptions.

I. Sustainability and Future Planning

(a) Ownership:

The IS team is fully capable of managing updates, troubleshooting issues, and providing staff support for both an online Google Site and on-premise software needs.

(b) Local Hosting:

This approach reduces reliance on purely cloud-based services for all functions, aligning with current security preferences and bandwidth realities. Google Sites secures core documents, while Google Forms introduces flexible, scalable workflows.

(c) Continuous Improvement:



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A Google Site serves as a valuable step in ROPSSA's and HCF's transition towards more integrated cloud-based software. Additionally, the switch to Office 2022 and Windows 11 software offers significantly more features than Office 2016 and Windows 10, including Copilot AI, which can be utilized regardless of our web platform.

VI. Conclusion

A. Summary of Final Report

This consulting engagement with the Republic of Palau Social Security Administration and the Healthcare Fund successfully addressed critical needs for modernizing internal documentation, optimizing workflows, and enhancing operational efficiency. The project focused on four key areas: updating employee and operations manuals, digitizing employee resources and form submissions, developing an AI-powered legal compliance and text synthesis tool, and advising executive leadership on workflow modernization.

B. Summary of Key Achievements

The project successfully delivered comprehensive solutions tailored to ROPSSA's and HCF's unique environment and future needs:

(a) Modernization of Employee Manual:

A comprehensive rewrite consolidated accurate policies from three disparate versions, eliminating contradictions, modernizing language, and reorganizing content for enhanced clarity and usability. This transformed a source of confusion into a unified, accurate reference for staff and managers alike.

(b) Development of Distinct Operations Manuals:



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Recognizing HCF's impending independence and the critical legal implications of Title 41 of the Palau National Code, two distinct, plain-language operations manuals were developed for ROPSSA and HCF. These manuals accurately reflect each entity's specific governance structure, operational processes, and future autonomy, mitigating legal risks and improving operational clarity.

(c) Creation of an AI-Powered Compliance and Drafting Tool:

An in-house AI tool was developed to automate legal compliance checks and provide text synthesis support for internal documents. This solution leveraged Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs) to reduce manual effort, ensure accuracy, and proactively identify compliance gaps, all within a secure, local environment.

(d) Workflow Modernization and Digital Transformation:

Through a detailed process audit and department consultations, executive leadership was advised on optimizing internal workflows. The recommended strategy focused on expanding the use of local LAN-hosted solutions, such as a website for digital manuals and online forms, to enhance efficiency and reduce reliance on paper-based processes.

C. Challenges and Solutions

Throughout the project, several challenges were encountered and successfully navigated:

(a) Outdated and Conflicting Documentation:

Initial plans for minor edits to existing manuals proved insufficient due to pervasive errors, outdated content, and structural disorganization across three versions. This necessitated a more comprehensive rewrite to ensure accuracy, consistency, and usability for staff.

(b) Technical Constraints and Internet Reliability:



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The limited internet connectivity in Palau and the organization's preference for local network operations due to security and cost concerns presented significant technical constraints. This influenced the decision to develop an in-house, locally hosted AI tool and to recommend a progressive workflow modernization strategy (LAN-based Website + Cloud Forms) rather than a full cloud migration.

(c) Hardware and Software Compatibility:

Ensuring the AI tool would run efficiently on existing hardware and integrating new digital solutions with the proprietary SQL database and IIS server required careful consideration and iterative development.

(d) Evolving Software Landscape:

The unexpected revelation that ROPSSA recently purchased licenses for Microsoft Office 2022 for 29 computers, after initial recommendations were made for a Microsoft 365 cloud-based package, necessitated a re-evaluation of the workflow modernization strategy. The final recommendation prioritized leveraging the existing Office 2022 investment and delaying a full cloud transition for 3-5 years, thus optimizing cost-effectiveness and alignment with current infrastructure.

D. Impact and Benefits

The implemented solutions are designed to deliver significant and sustainable benefits to ROPSSA and HCF:

(a) Enhanced Clarity and Consistency:

The updated manuals eliminated confusion, provided a unified reference, and streamlined onboarding for new hires, leading to more consistent policy application and improved staff confidence.

(b) Improved Operational Efficiency:



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By reducing manual, paper-based processes and duplicative data entry, staff time was redirected to more critical tasks, leading to increased productivity.

(c) Mitigated Legal and Compliance Risks:

The distinct operations manuals and the AI compliance tool ensured adherence to Title 41 of the Palau National Code, reducing the risk of legal non-compliance, financial penalties, and reputational damage, especially as HCF moves towards independence.

(d) Future-Proofing and Sustainability:

The locally hosted solutions, comprehensive training for the IS team, and designated policy owners ensured long-term sustainability and provided a clear upgrade path for future technology refreshes.

(e) Empowered Staff:

Providing easy access to accurate digital information and user-friendly tools minimized frustration, improved morale, and supported a culture of continuous improvement.

E. Acknowledgments

We are immensely grateful for the collaboration and support received from the Republic of Palau Social Security Administration and Healthcare Fund staff throughout this project. The feedback from management underscores the tangible impact of these solutions on daily operations, new member training, internal form submissions, employee resources, legal compliance, and text synthesis. We deeply appreciate the hospitality and invaluable experiences gained from our community partners and look forward to maintaining these close connections.



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F. Direct Feedback from Managers

"This project is incredibly important to us. Once finalized, the updated manual will help eliminate much of the confusion caused by years of amendments, administrative changes, and evolving laws without a unified reference. The work you and your colleague are doing will truly help us move forward with more clarity and consistency."

- **Shonya Kadol-Andreas, Employer Services Manager**

"I gave our new employee both the old manual and the new one and asked her if she had any thoughts. She said the new one is far more clear and she can actually understand what our policies are. That means a lot to us because it's coming from someone who doesn't already have knowledge of our system."

- **Pauline Reklai, Members Service Manager**

"Thank you both for your impressive work this summer. I really appreciate what you have done for our team and look forward to implementing insights from your work into our department. I wish you all the best and hope to stay in touch."

- **Jonathan Eberdong, Information Technology Manager**



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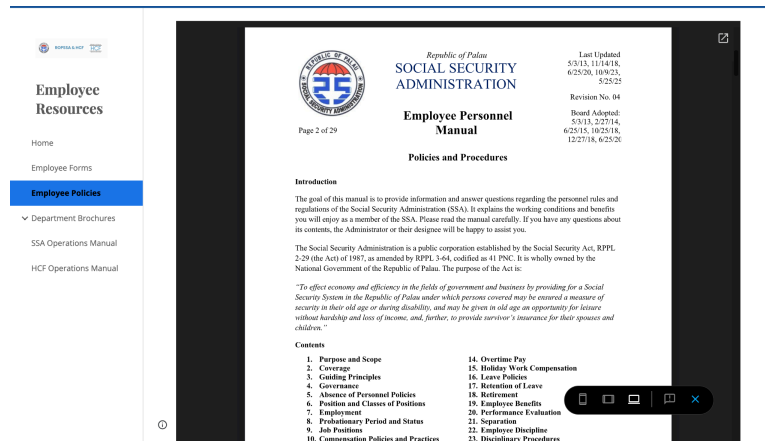
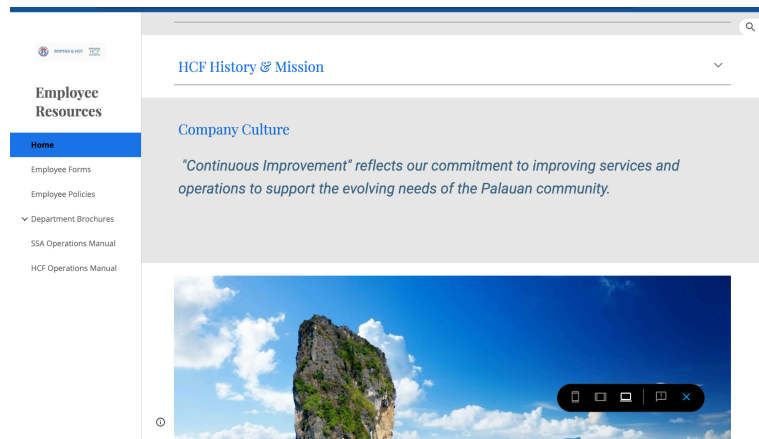
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VII. Appendices

A. Employee Resources Website





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B. Project Presentation

Modernizing Internal Workflows at ROPSSA and HCF

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June 2025

Why Now is the Time for Thoughtful Modernization

We believe now marks the time to consider new ways of modernizing and rethinking the way ROPSSA and HCF Operate. As they prepare for:

- The organizational separation of HCF into an independent entity
- The move to a new headquarters with upgraded hardware
- Ongoing improvements in service delivery and policy compliance

[Link to Presentation Slides](#)

C. AI Tool

[Troubleshooting Document](#)



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Documentation

